SÈRIE 1

Exercici 1

Q1 d

Q2 a

Q3 d

Q4 b

Q5 c

Exercici 2

a)

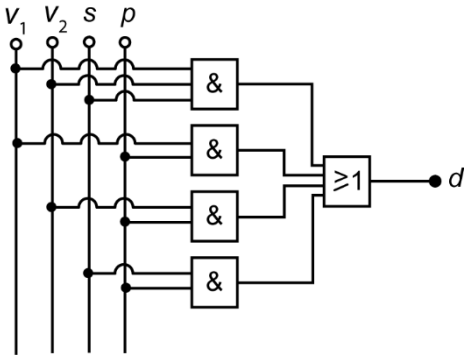
v_1	v_2	s	p	d
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

b)

$$d = \bar{v}_1\bar{v}_2sp + \bar{v}_1v_2\bar{s}p + \bar{v}_1v_2sp + v_1\bar{v}_2\bar{s}p + v_1\bar{v}_2sp + v_1v_2\bar{s}p + v_1v_2s\bar{p} + v_1v_2sp$$

$$d = v_1v_2s + sp + v_1p + v_2p$$

c)



Exercici 3

a)

$$P_{\text{mec}} = mgv = \frac{mg\left(h_2 - h_1\right)}{t}; \quad P_{\text{elèctr}} = \frac{P_{\text{mec}}}{\eta_{\text{mot}}} = 4,631\,\text{W}$$

b)

$$\omega_{\text{mot}} = \frac{v}{d/2} = 1,7\,\text{rad/s}; \quad \Gamma_{\text{mot}} = \frac{P_{\text{mec}}}{\omega_{\text{mot}}} = 2,452\,\text{Nm}$$

c)

$$\tilde{P}_{\text{elèctr}} = \frac{mg\left(h_2 - h_1\right)}{\eta_{\text{mot}}t/2} = 2\,P_{\text{elèctr}}$$

$$Inc = \frac{\tilde{P}_{\text{elèctr}} - P_{\text{elèctr}}}{P_{\text{elèctr}}} = 100\,\%$$

Exercici 4

a)

$$ce_{\text{pèl}} = \frac{c_{\text{pèl}}}{p_{c_\text{pèl}}} = 83{,}33 \times 10^{-3}\ €/\text{kWh}; \quad ce_{\text{gas}} = \frac{c_{\text{gas}}}{\rho\, p_{c_\text{gas}}} = 122{,}9 \times 10^{-3} €/\text{kWh}$$

b)

$$E_{\text{pèl}} = \frac{E_{\text{cons}}}{\eta_{\text{pèl}}}; \quad m_{\text{pèl}} = \frac{E_{\text{pèl}}}{p_{c_\text{pèl}}} = 3324\ \text{kg}$$

c)

$$E_{\text{gas}} = \frac{E_{\text{cons}}}{\eta_{\text{gas}}};\ V_{\text{gas}} = \frac{E_{\text{gas}}}{\rho\, p_{c_\text{gas}}} = 1729\ \text{L}$$

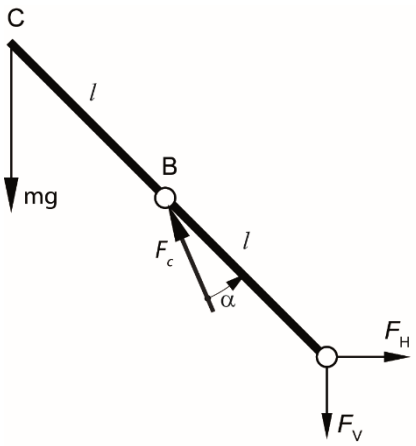
d)

diferència anual en el cost del combustible: $\Delta c_{\text{comb}} = V_{\text{gas}} c_{\text{gas}} - m_{pel} c_{\text{pel}}$

$$t = \frac{c_{\text{inv_pèl}} - c_{\text{inv_gas}}}{\Delta c_{\text{comb}}} = 5{,}444\ \text{anys}$$

Exercici 5

a)



b)

A partir del triangle OPB es pot veure que: $\varphi + 90 + 2\alpha = 180;\ \alpha = \dfrac{90-\varphi}{2}$

c)

$$\sum M(\mathrm{O}) = 0; \quad F_c \sin(\alpha)\, l - m\, g\, 2l \cos(\varphi) = 0;$$

$$\left.\begin{array}{l}\varphi = 30^{\mathrm{o}} \\ \alpha = 30^{\mathrm{o}}\end{array}\right\} \rightarrow \quad F_c = 2548\,\mathrm{N}$$

d)

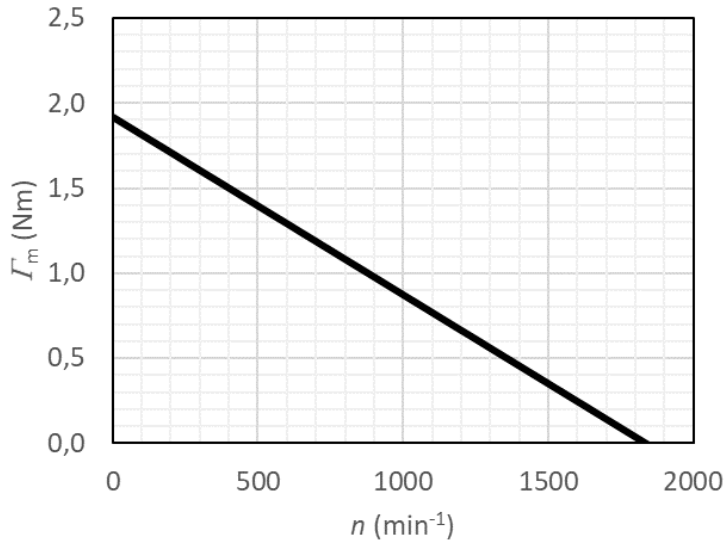
$$p_{\mathrm{int}} = \frac{F_c}{\frac{\pi d_{\mathrm{int}}^2}{4}} = 2{,}028\,\mathrm{MPa}$$

Exercici 6

a)

$$\tau = \frac{z_p}{z_r} = 0{,}1452$$

b)



La velocitat de gir és màxima quan el parell és nul: $n_{màx} = 192{,}0\,\text{rad/s} = 1833\,\text{min}^{-1}$

c)

$$\omega_{mot} = n_{mot}\frac{2\pi}{60}$$

$$I = \frac{P_{elèctr}}{U} = \frac{P_{mec}}{\eta U} = \frac{\Gamma_{mot}\omega_{mot}}{\eta U} = \frac{\left(0{,}08U - 0{,}01\omega_{mot}\right)\omega_{mot}}{\eta U} = 6{,}924\ \text{A}$$

d)

$$\omega_{con} = \tau\omega_{mot} = 15{,}20\ \text{rad/s}$$